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Experiment: 10

5G Network Slicing Environment Using MATLAB

OBJECTIVE 1 :

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
slices = {'eMBB','URLLC','mMTC','Private','IoT'};
```

```
count = [1 1 1 1 1]; % One instance of each network slice
```

```
figure
```

```
bar(count)
```

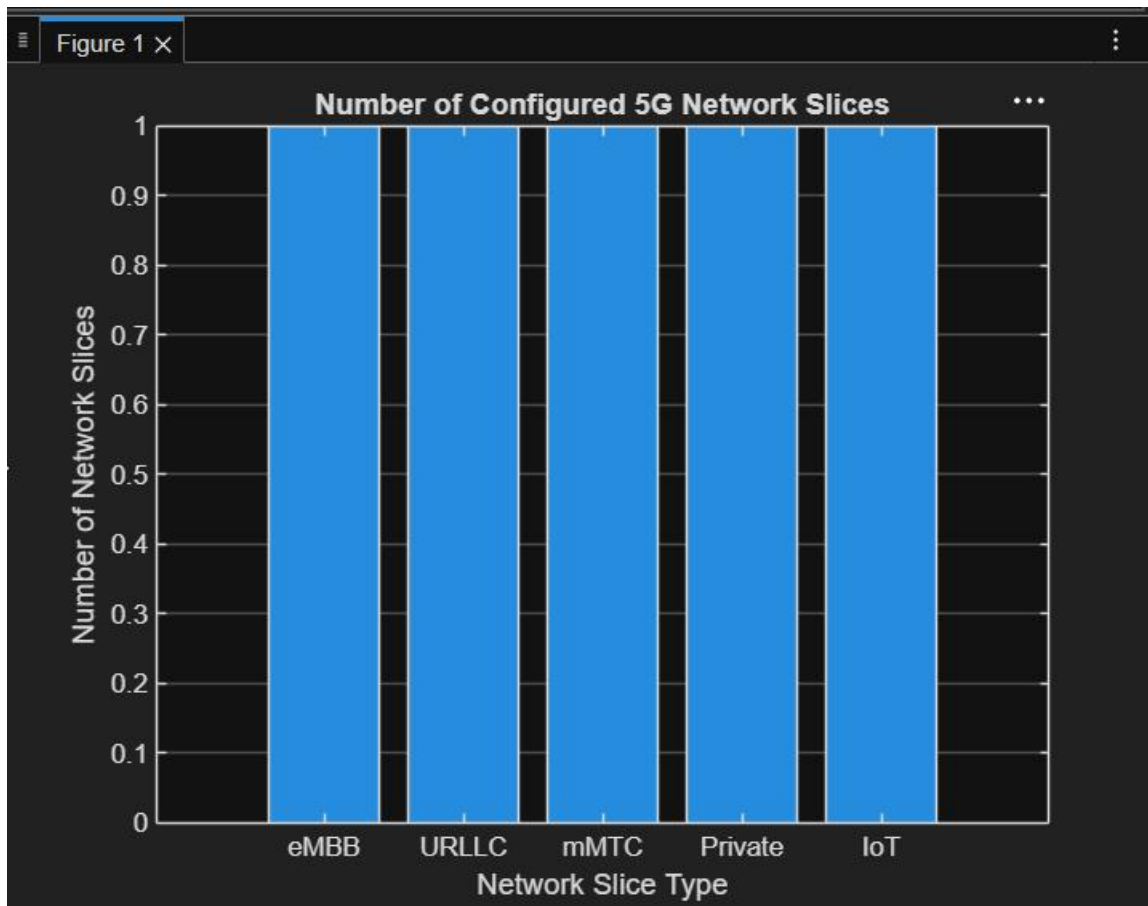
```
grid on
```

```
set(gca,'XTickLabel',slices)
```

```
xlabel('Network Slice Type')
```

```
ylabel('Number of Network Slices')
```

```
title('Number of Configured 5G Network Slices')
```



OBJECTIVE 2:

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});
```

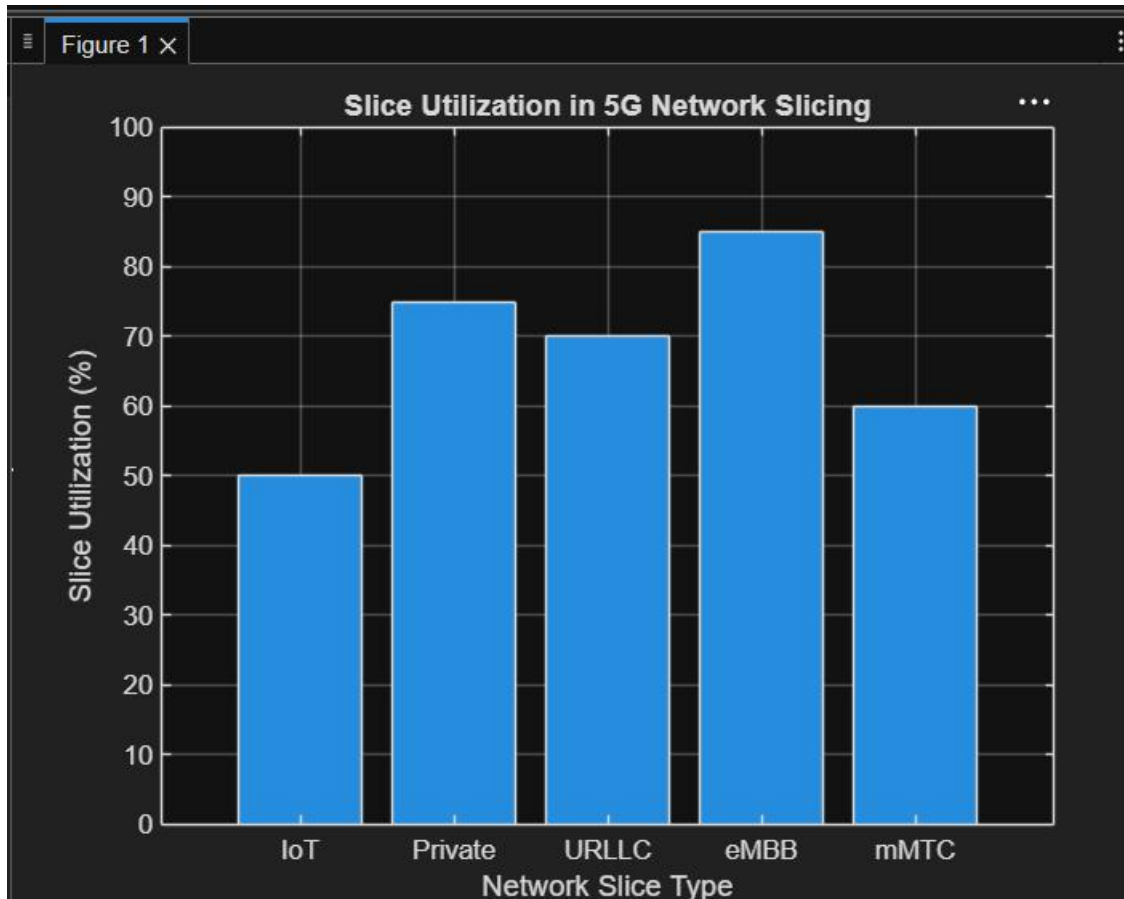
```
utilization = [85 70 60 75 50];
```

```
figure;
```

```
bar(slices, utilization);
```

```
grid on;
```

```
xlabel('Network Slice Type');  
ylabel('Slice Utilization (%)');  
title('Slice Utilization in 5G Network Slicing');  
ylim([0 100]);
```



OBJECTIVE 3:

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
slices = [1 2 3 4 5];  
latency = [25 5 40 15 60];  
  
figure;  
plot(slices, latency, '-o', 'LineWidth', 2, 'MarkerSize', 8);  
grid on;  
  
xticks(slices);  
xticklabels({'eMBB', 'URLLC', 'mMTC', 'Private', 'IoT'});  
  
xlabel('Network Slice Type');  
ylabel('End-to-End Latency (ms)');  
title('5G Network Slicing: End-to-End Latency');  
  
ylim([0 70]);
```

